

GenCC Submission Portal User Guide

Last Updated: Feb 2026

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A.1 Upload Errors

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Overview

The GenCC Submission portal facilitates the uploading, publishing, unpublishing, and republishing of submissions by pre-approved GenCC submitters. For more information on its features and use, please see the help documentation below and the submission portal demo recording [here](#). If you have any questions related to the functionality of the portal, please email gencc-tech@thegencc.org. If you have any questions related to the GenCC and/or scientific content, please email gencc@thegencc.org.

The submission portal can be found at <https://portal.thegencc.org/>. Users will be required to log in using their registered email as their username and password. A GenCC submitter, once pre-approved, will be registered for the GenCC portal and an account will be created for both the submitter and at least one member/contact.

GenCC Glossary

Submission - A single GenCC record created by a submitter, defined by a specific **gene**, **disease**, and **mode of inheritance**.

Publish - Making a submission publicly available by adding it to the production (live) GenCC database.

Unpublish - Removing a submission from the public GenCC database so it is no longer visible to users.

Republish - Publishing an updated version of a previously published submission, typically after making changes (e.g. clinical validity strength, assertion criteria, comment on clinical significance)

Job - A queued request to perform one or more actions on one or more submissions (for example: publish, unpublish, or republish). Jobs are processed and applied to the live database during a release.

SGC ID - A unique versioned identifier for a Submission. The version of this ID increases with each action taken on a submission (including unpublish). This ID is required for unpublishing a submission.

Release - A batch update that applies all processed jobs to the publicly available GenCC data. Releases are typically run weekly. A new release is only created if at least one job has been submitted since the previous release.

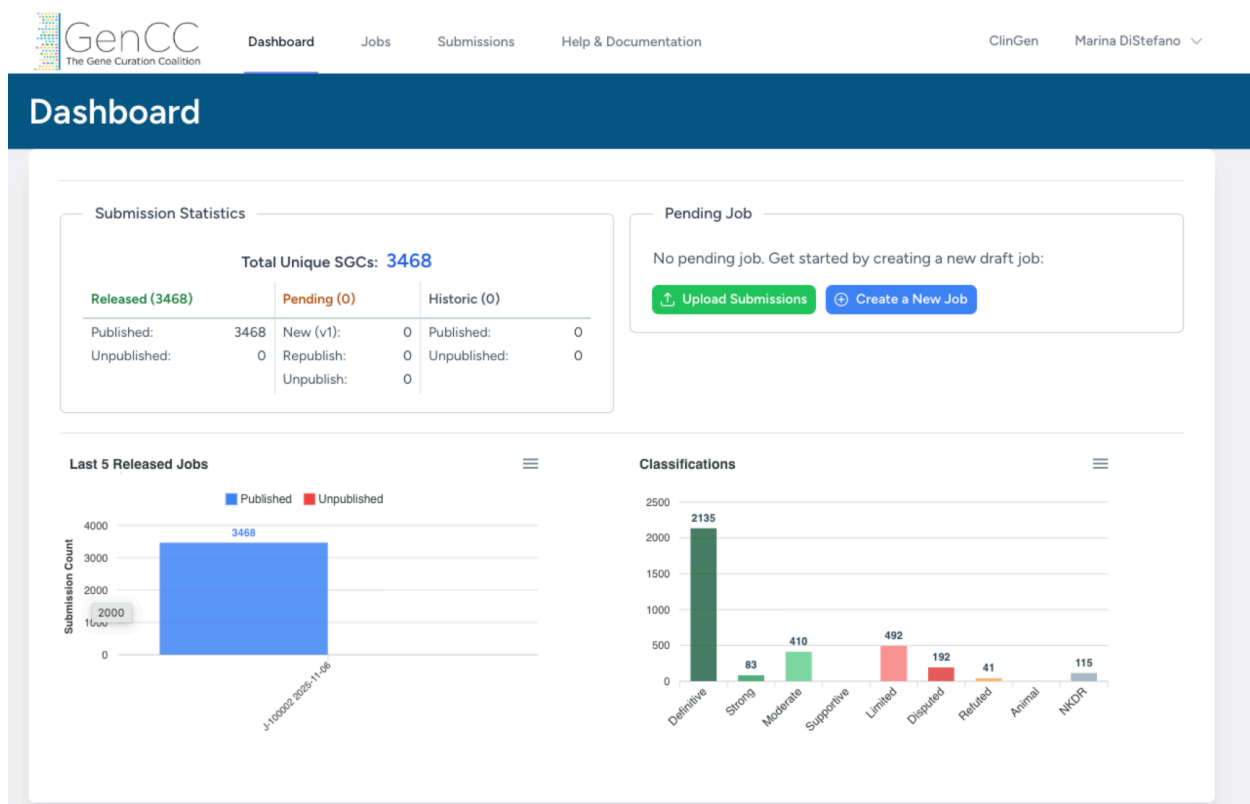
Pending - Submission actions that are not yet live. Pending actions are associated with **Draft** or **Submitted** jobs. Once a job is published and included in a release, its actions are no longer pending.

Live - The current, authoritative state of a submission. Live submissions represent the **most recent version** for each SGC ID. A submission can be live whether it is currently **published** or **unpublished**. Live submissions are not pending.

Historic - Older versions of a submission that are no longer current. Any published or unpublished version of an SGC ID that is **not the most recent version** is considered historic. Historic records are retained for tracking and audit purposes but are not authoritative.

GenCC Dashboard

Below is a screenshot of the Dashboard (with ClinGen used as an example submitter)



The clickable sections at the top are as follows:

Dashboard - takes you back to this main dashboard view

Jobs - takes you to a queue of all of your current and previous jobs with the ability to view, sort, filter, search, or download them. See the executing jobs section of this help documentation for further information.

Submissions - takes you to a list of all active submissions with the ability to view, sort, filter, search, download or take action and move them to jobs. See the submissions sections of this help documentation for further information

Help and documentation - takes you to a page with all of the help documents. This pages includes this help document, the GenCC submission directions and spreadsheet download, and a Contact Us section

The submitter organization is listed at the top right (e.g. ClinGen) and is not clickable. However, clicking on the name of the person logged in (e.g. Marina DiStefano) activates a drop down where the user can edit their profile and the submitter profile, including all of the submitter details that are viewable on the GenCC public database (i.e. Logo, description, website, assertion criteria, and public contact).

The visualizations on the dashboard are as follows:

Submission statistics: This provides a table overview of all submissions for that submitter. It includes the number of unique submissions and then sorts them by status. The left column of the table details released submissions (both published and unpublished), the center column details pending (not yet live in the database) submissions and splits them by new, republished, and unpublished. The historic section shows prior published and unpublished submissions.

Pending job: This section provides the status of a pending job. Statuses include draft, submitted, and processed. If no job is pending, it is a place where users can start a job by either uploading a bulk submission with a GenCC Submission spreadsheet (Upload submissions button) or by creating a new job and manually adding submissions (create a new job button).

Last 5 released jobs: This section shows a bar graph of published and unpublished submissions for the most recent 5 jobs labeled by job. The three horizontal bars in the top right are clickable and allow downloading this data in different file formats (SVG, PNG, and CSV).

Classifications: This section shows a bar graph of the clinical validity of all published classifications (submissions). The three horizontal bars in the top right are clickable and allow downloading this data in different file formats (SVG, PNG, and CSV).

Executing Jobs

Jobs can be started in multiple locations and submissions can be added either manually or in bulk using the GenCC submission spreadsheet. See below for instructions on both. To start a

job from the Dashboard, the user can click the “upload submissions” or the “create a new job” buttons. The same buttons are also available on the Jobs tab.

Jobs tab: Below is a screenshot of the Jobs tab (with ClinGen used as an example submitter)

The screenshot displays the 'Jobs' tab in the GenCC Submission Portal. At the top, there are two buttons: 'Upload Submissions' and 'Create a New Job'. Below these is a yellow banner with the text 'Create a new job or upload submissions to get started.' The main content area features a table of jobs. The table has the following columns: Job Id, # Submissions, Created By, Status Date, Status, and Action. A single job is listed with the ID J-100002, 3468 submissions, created by Marina DiStefano, a status date of 2025-11-06, and a status of 'Released'. The job is marked as a favorite with a star icon. The Action column contains a green arrow icon. Above the table, there is a 'Download' button, a 'Total of 1 jobs.' label, a 'Show All' dropdown, a 'Submitter' dropdown, and a 'Keyword Search' input field. At the bottom, there is a pagination bar showing 'Total of 1 jobs.' and a page number '1'.

Job Id	# Submissions	Created By	Status Date	Status	Action
J-100002 GenCC Import 2026-01-02	3468	Marina DiStefano	2025-11-06	Released	→

This tab lists all jobs with their statuses. Possible statuses are Draft, Pending, Submitted, and Released. Jobs can be downloaded using the green download button, filtered by status using the dropdown button that says “show all” in this screenshot, and searched by keyword. If there are multiple approved users for a GenCC submitter the “submitter” button can be toggled to show all submissions for a GenCC submitter or those only done by the current logged in user (“User only”).

Jobs can be marked as a favorite by clicking the star button on the far left of the job. This can then be used for filtering. To view all submissions in a particular job, click the arrow under the Action column. See the screenshot below for an example of a particular job.

Job: J-100002
GenCC Import 2026-01-02

← Return to Job List

Job has been released.

Submitter:

ClinGen
GENCC:000102

Submitter by:

Marina DiStefano
mdistefa@broadinstitute.org

Status Date:

2025-11-06

Status:

Released

Download

Total of 3468 submissions.

Show All

Historic

Submitter

Keyword Search

☐	Submission ↑↓	Gene ↑↓	Disease ↑↓	Inheritance ↑↓	Classification ↑↓	Status Date ↑↓	Status ↑↓	Action
☐	SGC-104217.1 J-100002	PTCSC1 HGNC:37127	hereditary nonpolyposis colon cancer MONDO:0018630	Autosomal dominant HP:0000006	Limited GENCC:100004	2025-12-05	Published	☐ <input type="star"/> <input type="refresh"/> <input type="eye-slash"/> <input type="arrow-right"/>
☐	SGC-104218.1 J-100002	PNRC1-DT HGNC:55238	myopathy caused by variation in POMGNT1 MONDO:0700068	Autosomal recessive HP:0000007	Definitive GENCC:100001	2025-12-05	Published	☐ <input type="star"/> <input type="refresh"/> <input type="eye-slash"/> <input type="arrow-right"/>
☐	SGC-104219.1 J-100002	MRT04 HGNC:18477	mitochondrial disease MONDO:0044970	Mitochondrial HP:0001427	Limited GENCC:100004	2025-12-05	Published	☐ <input type="star"/> <input type="refresh"/> <input type="eye-slash"/> <input type="arrow-right"/>

The job ID is listed at the top, along with the status, status date, and submitter. Much like the jobs page, this page can be downloaded, filtered and keyword searched. It has a clickable submitter toggle and a “hide historic submissions button” that will hide older inactive versions of submissions.

Each submission row can be marked as a favorite by clicking the star. There are multiple buttons that can be clicked in the Action section. The circular blue arrow is the republish button. The red eye with a strikethrough is the unpublish button, and the green arrow is the view button. **Of note, clicking the republish or unpublish buttons will move the new submission to a new or pending job.** These actions can also be performed in bulk by choosing the select box at the far left of the job and choosing the action from the top yellow action bar that pops up (see screenshot below). For more information on editing submissions, please see the submission sections of this help document.

2 submission(s) selected

Download

Total of 3468 submissions.

Show All

Historic

Submitter

Keyword Search

☐	Submission ↑↓	Gene ↑↓	Disease ↑↓	Inheritance ↑↓	Classification ↑↓	Status Date ↑↓	Status ↑↓	Action
☒	SGC-104217.1 J-100002	PTCSC1 HGNC:37127	hereditary nonpolyposis colon cancer MONDO:0018630	Autosomal dominant HP:0000006	Limited GENCC:100004	2025-12-05	Published	☐ <input type="star"/> <input type="refresh"/> <input type="eye-slash"/> <input type="arrow-right"/>
☒	SGC-104218.1 J-100002	PNRC1-DT HGNC:55238	myopathy caused by variation in POMGNT1 MONDO:0700068	Autosomal recessive HP:0000007	Definitive GENCC:100001	2025-12-05	Published	☐ <input type="star"/> <input type="refresh"/> <input type="eye-slash"/> <input type="arrow-right"/>

Uploading Submissions from a Spreadsheet

To upload Jobs from a spreadsheet, the submitter should prepare the GenCC submission spreadsheet ahead of time. This sheet and instructions for using it can be found [here](#).

****Please note that this submission spreadsheet is now in Version 2 and has changed from the original sheet. The two columns that have been added are “sgc_id” and action. SGC ID is required for updates only (republishing, unpublishing). All submissions have been assigned an SGC ID. Please refer to the submissions tab to find these IDs via searching or through bulk download. In the action column (column B), submitters must designate if a submission is new, republished, or unpublished.**

Once this sheet has been completed with the desired submissions, it can be uploaded using the upload submissions button on the Dashboard or the Jobs page. Clicking this button will automatically start a new job if no job is open or add to your current job. **Please note that only one file can be uploaded per job.** Choose the file you would like to upload and click the open button. When a file is chosen, a box will pop up with the status of the file validation. When the file has completed validation, the submissions with the relevant errors (if applicable) will be displayed. See Appendix A for error modes.

To correct a file and re-upload it, you can download the file from the Job page or correct your original file. The original file with errors must be deleted using the trashcan icon. An image of a job with file upload errors is provided below:

The screenshot displays the 'Job: J-100022' page. At the top right is a 'Return to Job List' link. A yellow banner states 'There are submission errors present'. Submission details include: Submitter (ClinGen, GENCC:000102), Submitter by (Marina DiStefano, mdistefa@broadinstitute.org), Status Date (2026-02-17), and Status (Draft). The 'Uploaded File' section shows 'GenCC Submission Spreadsheet.xlsx' (75.36 KB, Uploaded 2 minutes ago) with an 'Invalid' status and a trash icon. A red-bordered box contains a warning: '1 Data Validation Error(s) - GenCC Submission Spreadsheet.xlsx', with 'Download' and 'Hide' buttons. A yellow note at the bottom says: 'Please review and correct the errors below. For detailed guidance on submission requirements, refer to GenCC Submission Directions.' with an external link icon.

Adding Submissions Manually in the Portal

Submissions can be added via spreadsheet upload (see instructions above) or manually in the submission portal. If there is no open job, manual entries must be made by clicking the “+ Create a New Job” button on the dashboard or the Jobs tab.

If there is currently an open job, as in the screenshot above, clicking the “+New” button will create a new submission. Clicking this button will open a pop up box where the user will need to

attest that they want to create a new submission. While only one file can be added to a job, a user is not limited in the number of manual submissions added to a job.

A manual submission screen will then open and users will need to enter all required fields manually to get rid of submission errors. See the screenshot below. To edit the required fields, click the red X on the far right of the row. This will open up a dropdown screen for fields with standardized entries (e.g. Classification and Mode of Inheritance) or a free text entry for others.

This submission has errors
Fields with errors are highlighted below in red. Click on the field edit button to correct. For a submission to be published to GenCC, all errors must be resolved. Once all errors have been cleared, the system will automatically stage the submission for publishing.

Draft Submission (New). Make edits below or delete this submission.

Submitter:	ClinGen GENCC:000102	Submitted by:	Marina DiStefano mdistefa@broadinstitute.org
Status Date:	2026-02-17	Status:	New

Local Key:		
Classification:	-	
Gene:	-	
Disease:	- -	
Mode Of Inheritance:	-	
PubMed IDs:		
Public Report:		
	Evaluated Date:	
Assertion Criteria:		
	Name:	
Notes Public:		
	Private:	
Primary Contributor:		
Description of Change:		
Workflow:	creation_date : 2026-02-17T15:24:00.955764Z	

Unpublishing submissions

Submissions can be unpublished via the Submission spreadsheet or manually. **Of note, unpublishing a submission will increase its version number and remove it from the public GenCC database.**

For unpublishing via the submission spreadsheet, the SGC ID of the submission should be included in column A and “U” should be included in column B for the action, which corresponds to unpublish. **All other columns MUST be empty.**

For unpublishing manually in the submission portal, submissions can be accessed on the Submissions tab. On this tab, submissions can be filtered, downloaded, keyword searched, or favorited, much like on the Jobs tab. To unpublish a particular submission, click the red eye with a strikethrough. A pop up box will appear asking the user to confirm that they want to unpublish the submission. Clicking yes will add this submission to either an already open job or create a new job.

Republishing submissions

Submissions can be republished via the Submission spreadsheet or manually. **Of note, republishing a submission will increase its version number and should only be done if a user plans to change a fundamental aspect of the record.** For republishing via the submission spreadsheet, the SGC ID of the submission should be included, “R” should be included in column B for the action, which corresponds to republish, and all required fields must be completed.

For republishing manually in the submission portal, submissions can be accessed on the Submissions tab. On this tab, submissions can be filtered, downloaded, keyword searched, or favorited, much like on the Jobs tab. To republish a particular submission, click the circular blue arrow. A pop up box will appear asking the user to confirm that they want to edit a currently published submission (if you are editing one that is live) or republish an unpublished submission (if you are editing one that is unpublished). Clicking yes will add this submission to either an already open job or create a new job.

Completing a Job

When a submitter has completed adding submissions to a job and all errors have been resolved, they can submit the job. **Of note, when a job has been submitted, it can no longer be edited and will be queued for submission.** Jobs are processed weekly on Saturday at midnight. A user cannot submit another job while a previous one is in a pending state.

Once all errors have been resolved from a job the banner at the top will turn yellow with the message “There are no errors.” (see image below). The submit button at the far right can now be clicked. Once this is clicked, a pop will appear and the user must confirm that they want to submit the job.

Job: J-100023 [Add a Label](#) ← Return to Job List

There are no errors. Submit

Submitter: ClinGen
GENCC:000102

Submitter by: Marina DiStefano
mdistefa@broadinstitute.org + New

Status Date: 2026-02-17

Status: Draft Upload

The user can also submit a job from the Dashboard by clicking the submit button in the Pending Job section. See screenshot below:

Dashboard GCI Sync Submissions

Submission Statistics

Total Unique SGCs: **3468**

Released (3468)		Pending (1)		Historic (0)	
Published:	3468	New (v1):	0	Published:	0
Unpublished:	0	Republish:	1	Unpublished:	0
		Unpublish:	0		

Pending Job

Job: J-100024 New (v1): 0 Submit

Status: draft Republish: 1

Date: 2026-02-17 Unpublish: 0

When a job has been successfully submitted, users can check the status in this Pending job section on the Dashboard or in the Jobs tab. If you notice an error in your submitted job, you will need to wait for the job to process before correcting the error in a new job.

Frequently Asked Questions:

This list will be updated as common questions are fielded over time. If your question does not appear here, please contact gencc-tech@thegencc.org.

How do I change submitter details for my organization? This can be done in the user profile section of the submission portal. From the dashboard, click your name in the top right corner. A drop down menu will pop up. Click the submitter button. This will take you to the submitter profile where you can edit details such as Logo, website, and submitter description.

Why are my new submissions not live on the GenCC production database? Jobs are processed weekly on Saturdays at midnight so users will not see their submissions go live until that weekly job processing is completed.

How do I edit a submitted job? Once jobs are submitted, they cannot be edited. If a user notices an error in a previously submitted job, they can correct the error in a new job once the old job is processed.

How do I add multiple files to a job? A job can only have one file upload. If you want to add multiple files to a job, they will need to be combined into one submission sheet. Please delete the current file associated with the job and upload your new combined file.

Why do I need to reference SGC IDs for unpublished and republished submissions? SGC IDs are unique versioned identifiers for submissions. They are required in order for the database to correctly identify and version a record and are now publicly displayed in the database for reference purposes.

How do I find an SGC ID? All existing submissions were assigned SGC IDs. These can be located both individually (by searching) and in bulk (by download) in the submissions tab of the submission portal.

Appendix A

A.1 Upload Errors

These occur during the Uploading and Validation process before the background processing begins. NOTE: you only see the background processing for files that have a large amount of records. See A.2 for Background processing Errors

Error Type	User Title	Message Template	Scenario
6003	Error	"File upload failed. The file may be too large. Maximum upload size is 20MB. Please contact support if you need to upload larger files."	File exceeds PHP's <code>upload_max_filesize</code> or <code>post_max_size</code> limits
invalid_file_format	Invalid File Format	"The uploaded file does not appear to be a valid GenCC submission template. Please download the official template from the GenCC website and ensure your data starts on row 13."	File has fewer than 13 rows (data expected to start at row 13)
missing_header_row	Missing Header Row	"The uploaded file is missing the required header row. Headers must be in row 6 of the spreadsheet. Please use the official GenCC submission template."	Row 6 is empty or null
invalid_header_columns	Invalid Column Headers	"The file headers do not match the expected GenCC submission template. Missing columns: {list}. Unexpected columns: {list}. Please download and use the official template from the GenCC website."	Headers don't match expected columns at row 6

A.1.1 Required Field Errors

Error Type	Message Template	Scenario
missing_required_field	"Required field '{field}' is missing."	A required column is empty for a data row

Required Fields:

- **action** - Must be N, R, or U
- **hgnc_id** - Gene identifier
- **disease_id** - Disease identifier
- **moi_id** - Mode of inheritance
- **submitter_id** - Submitter organization
- **classification_id** - Classification level
- **date** - Report/evaluation date
- **assertion_criteria_url** - Link to assertion criteria

Note: For Unpublish (U) action, only **sgc_id** and **action** are required.

A.1.2 Field Format Errors

Error Type	Column	Message Template	Valid Format	Scenario
invalid_field_format	sgc_id	"Invalid format for column 'sgc_id'"	SGC-##### or empty	SGC ID doesn't match pattern
invalid_field_format	action	"Invalid format for column 'action'. Must be one of: N (New), R (Republish), or U (Unpublish)."	N, R, or U (case-insensitive)	Invalid action code
invalid_field_format	hgnc_id	"Invalid format for column 'hgnc_id'. Must be a valid HGNC gene identifier (e.g., HGNC:1234). Verify at https://www.genenames.org/ "	#### or HGNC:####	Gene ID doesn't match numeric or HGNC: format
invalid_field_format	disease_id	"Invalid format for column 'disease_id'. Must be a valid disease identifier from MONDO, OMIM, ORPHA, or Orphanet (e.g., MONDO:0001234, ORPHA:90044)."	MONDO:####, OMIM:####, ORPHA:####, Orphanet:####	Disease ID format invalid
invalid_field_format	date	"Invalid date format for column 'date'. Expected format: YYYY-MM-DD (e.g., 2024-01-15). Got: '{value}'"	YYYY-MM-DD or YYYY/MM/DD	Date can't be parsed

invalid_field_format	public_report_url	"Invalid format for column 'public_report_url'. Must be a valid URL starting with http:// or https://"	Empty or http://... / https://...	URL doesn't start with http(s)://
invalid_field_format	assertion_criteria_url	"Invalid format for column 'assertion_criteria_url'. Must be a valid URL to the assertion criteria documentation."	http://... or https://...	URL doesn't start with http(s)://
invalid_field_format	pmids	"Invalid format for column 'pmids'"	Comma/semicolon-separated numbers	PMIDs contain non-numeric characters

A.1.3 Field Value Errors (Database Lookups)

Error Type	Column	Message Template	Scenario
invalid_field_value	hgnc_id	"Invalid value for column 'hgnc_id'. Must be a valid HGNC gene identifier (e.g., HGNC:1234). Verify at https://www.genenames.org/ "	Gene ID not found in genes table
invalid_field_value	disease_id	"Invalid value for column 'disease_id'. Must be a valid disease identifier from MONDO, OMIM, ORPHA, or Orphanet"	Disease ID not found in diseases table or has no MONDO mapping
invalid_field_value	moi_id	"Invalid value for column 'moi_id'. Must be a valid HPO mode of inheritance term (e.g., HP:000006 for Autosomal dominant)."	MOI CURIE not in inheritances table
invalid_field_value	classification_id	"Invalid value for column 'classification_id'. Must be a valid GenCC classification (e.g., GENCC:100001 for Definitive)."	Classification CURIE not in classifications table
invalid_field_value	submitter_id	"Invalid value for column 'submitter_id'"	Submitter CURIE doesn't match the user's organization

A.1.4 Action-Specific Errors

Error Type	Message Template	Scenario
new_with_sgc_id	"Action 'N' (New) cannot have an SGC_ID. The SGC_ID column must be empty for new submissions."	Action=N but SGC_ID column is populated
action_misusing_sgc_id	"Action '{action}' ({name}) requires a valid SGC_ID in the first column."	Action=R or U but SGC_ID is empty

invalid_sgc_id_format	"Invalid SGC ID format. Must be 'SGC-' followed by exactly 6 digits with no leading zeros (e.g., SGC-100001)."	SGC ID doesn't match SGC-1##### pattern
unpublish_has_data	"Action 'U' (Unpublish) should only have SGC_ID and Action columns filled. The following fields must be empty: {fields}."	Unpublish row has data in fields other than SGC_ID and Action

A.1.5 SGC ID State Validation Errors

Error Type	Message Template	Scenario
sgc_id_not_found	"SGC ID not found. Only previously processed submissions can be republished or unpublished."	SGC ID doesn't exist in database OR belongs to different submitter
sgc_id_in_draft_or_submitted_job	"SGC ID is already in another draft or submitted job. Complete or cancel that job first."	Trying to R/U a submission that already has a pending draft/submitted version
republish_invalid_state	"Action 'R' (Republish) requires SGC ID to be in 'Published' or 'Unpublished' state. Check the current state of the submission."	Trying to republish a submission that isn't published or unpublished
unpublish_invalid_state	"Action 'U' (Unpublish) requires SGC ID to be in 'Published' state. Check the current state of the submission."	Trying to unpublish a submission that isn't currently published
republish_gene_change	"Action 'R' (Republish) cannot change the gene of an existing submission. To submit a different gene-disease association, create a new submission instead."	Republish row has different HGNC ID than original submission

A.1.6 Duplicate Detection Errors

Error Type	Severity	Message Template	Scenario
unique_column_requirement	Error	"Unique column requirement on column 'sgc_id' not met."	Same SGC_ID appears multiple times in spreadsheet
duplicate_sgc_id	Error	"Duplicate SGC ID found in spreadsheet. Each SGC ID must appear only once."	Repeated SGC_ID in file
duplicate_submission (blocking)	Error	"Duplicate submission found. A submission with the same gene, disease, and mode of inheritance already exists ({sid}, status: {status}). Each submitter may only have one active submission per gene-disease-MOI combination."	Gene-Disease-MOI combination already exists as published or draft

duplicate_submission (batch)	Error	"Duplicate rows within file: ({rows}), ({rows}). Each group contains rows with the same gene, disease, and mode of inheritance combination."	Multiple rows in same file have identical Gene-Disease-MOI
unpublished_duplicate_submission	Warning	"An unpublished submission exists with the same gene, disease, and mode of inheritance ({sgc_ids}). You may proceed, but consider republishing the existing submission instead of creating a new one."	Gene-Disease-MOI exists but is unpublished (allows upload to proceed)

A.1.7 PMID Errors

Error Type	Message Template	Scenario
invalid_pm_id_format	"Invalid PMID format found: Must be numeric with no leading zeros."	PMID contains non-numeric characters or has leading zeros (e.g., "0123456")
duplicate_pmids	"Duplicate PMIDs found in single submission: {pmids}. Each PMID should appear only once per submission."	Same PMID listed multiple times in one row

A.1.8 Empty File Error

Error Type	Message	Scenario
validation_error	"File contains no valid submission rows"	File has correct headers but no data rows (all rows empty)

A.2 Processing Phase Errors

These occur during asynchronous background processing.

A.2.1 Timeout/Termination Errors

Error Type	Severity	Message Template	Scenario
partial_upload	Warning	"Upload timed out after processing {n} of {total} submissions. The upload process exceeded the time limit and was stopped. The remaining submissions will not be included when submitting this job. Splitting your file into smaller batches is recommended for the remaining submissions."	Processing exceeded wall-clock timeout (default 10 minutes)

<code>partial_upload</code>	Warning	"Upload was interrupted after processing {n} of {total} submissions. The process may have timed out or been terminated unexpectedly. The remaining submissions will not be included when submitting this job. Splitting your file into smaller batches is recommended for the remaining submissions."	PHP process terminated unexpectedly (memory, queue restart, etc.)
<code>partial_upload</code>	Error	"Upload failed due to a fatal error after processing {n} of {total} submissions. Error: {message}. Please check your file and try uploading again."	PHP fatal error during processing (E_ERROR, E_PARSE, etc.)

A.2.2 Row Processing Errors

Status	Message Template	Scenario
<code>header_error</code>	"Error: Action {action} requires SGC_ID. Row:{row}"	R/U action missing SGC_ID (should be caught in validation, fallback)
<code>header_error</code>	"Error: SGC ID {sgc_id} not found! Row:{row}"	SGC_ID not in database (should be caught in validation, fallback)
<code>header_error</code>	"Error: Failed to process row {row}. Invalid data format."	JSON template rendering failed
<code>header_error</code>	"SGC ID {sid} encountered submission errors. Row: {row}"	<code>load_from_json()</code> returned validation errors

A.2.3 Submission-Level Validation Errors (load_from_json)

These are stored in `submission_errors` JSON field and will show on the submission UI:

Error Key	Message	Scenario
<code>gene_hgnc_id</code>	"Invalid HGNC ID"	Gene lookup failed (uses placeholder "-" gene)
<code>disease_curie_id</code>	"Invalid Disease ID"	Original disease CURIE not found (uses MONDO:0000001 fallback)
<code>disease_curie_id</code>	"Invalid Disease ID - no MONDO mapping found"	Disease exists but has no MONDO normalization mapping
<code>moi_curie_id</code>	"Invalid MOI ID"	MOI lookup failed (uses HP:0000005 "Unknown" fallback)
<code>classification_curie_id</code>	"Invalid Classification ID"	Classification lookup failed (remains null)
<code>report_date</code>	"Missing Report Date"	Report date field is empty
<code>report_date</code>	"Invalid Report Date"	Report date can't be parsed by Carbon
<code>report_url</code>	"Invalid Report URL"	URL fails <code>filter_var(FILTER_VALIDATE_URL)</code>

invalid_pmid	"Invalid PMID format: must be numeric"	PMID contains non-numeric characters
invalid_pmid	"Invalid PMID format: no leading zeros allowed"	Multi-digit PMID starts with 0